



Use a positive test charge to test the point where electric force is zero.

Since the resultant electric force between the charges is non-zero, pointing to the right - eliminate A and B.

On the right side, the positive test charge experiences a rightward repulsive force due to $+2Q$ and a leftward attractive force due to $-Q$. Let the neutral point be distance x from $-Q$.

$$\frac{1}{4\pi\epsilon_0} \left(\frac{2Q}{(1+x)^2} + \frac{-Q}{x^2} \right) = 0$$

$$2x^2 = (1+x)^2$$

$$x^2 - 2x - 1 = 0$$

$$x = 2.4 \text{ units}$$

Alternatively, test position C:

$$E \text{ due to } +2Q \propto 2Q / 2^2 = Q/2$$

$$E \text{ due to } -Q \propto Q / 1^2 = Q$$

Thus they can't cancel one another.

Thus, the best answer is D.